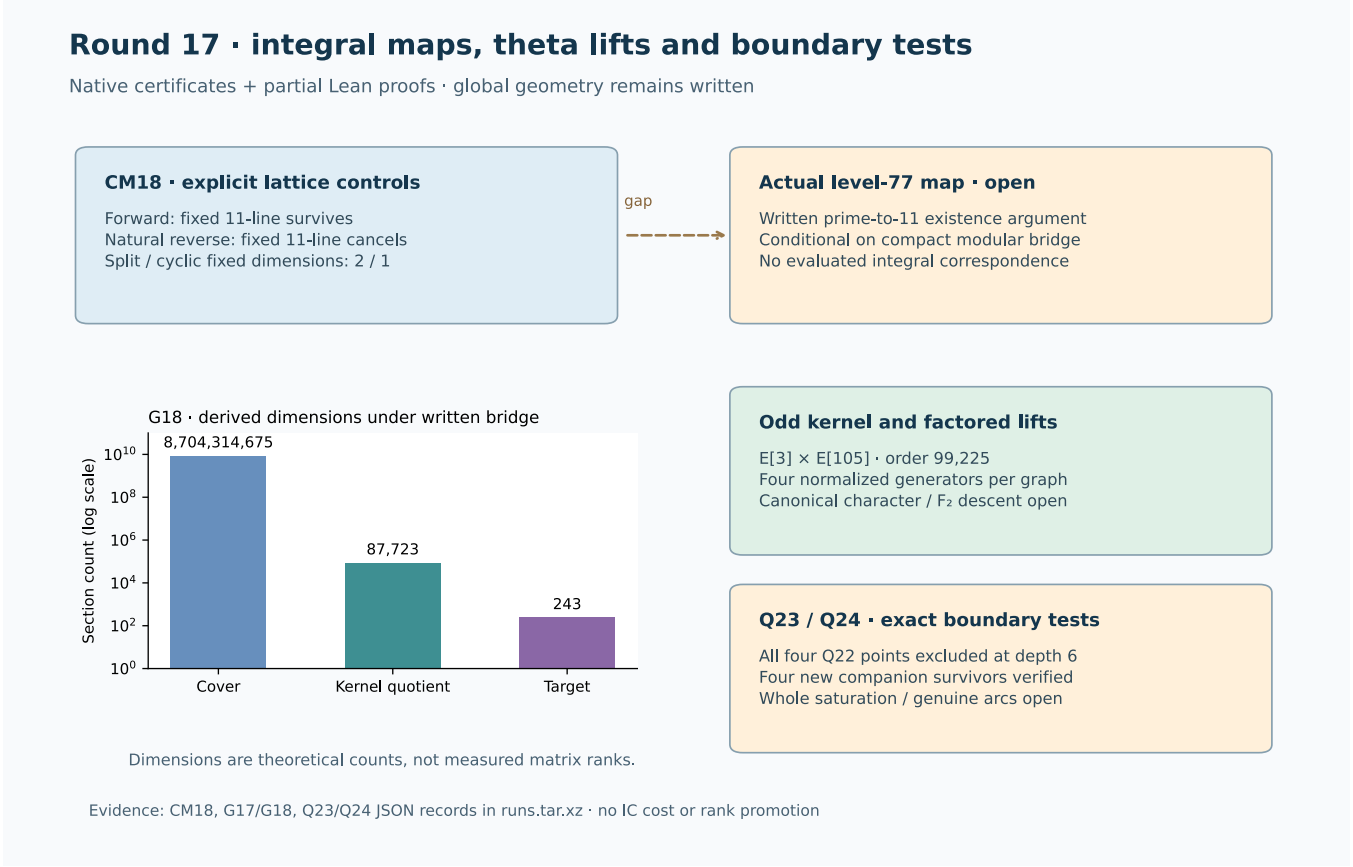
Round17: integral map existence, odd theta lifts and survivor exclusions

Client date22 October2026, America/Los_Angeles; execution23 October UTC. Scientific parent dcc9802e717465708a3e69562f4498bd7046def4. The base merge preserves all earlier evidence and both problem-index entries. This round continues the three requested tracks with exact native experiments and relevant Lean proofs. No IC timing, rank gain or efficiency verdict is supplied.

The main new lead is a written prime-to11 injection existence argument for the level77 carrier, alongside explicit integral lattice controls. The odd cover now has actual kernel generators and factored normalized theta lifts. All four Q22 survivors are excluded by source-bound quartics; four new Q24 survivors keep full saturation open. Canonical character matching, target section extraction and an evaluated modular correspondence remain open.



Integral level77 map: what is established

CM18 checks that each actual degree8 n=5 source Frobenius factor has eight simple roots modulo11. The trace-zero augmentation matrix has a cyclic basis of determinant-64 for both signs, giving an isomorphism with its companion lattice at11. An integral16-by8 intertwiner into the cyclic mixed-sign lattice preserves the fixed11 line. The natural reverse map kills that line. Generated Lean proves the actual intertwiner, fixed source, nonzero image, reverse cancellation, root counts and derivative checks.

Two integral controls with the same Frobenius polynomial and det(F-1)=121 have different11-primary cokernels: the split control gives(11,11), the cyclic glued control gives(121). Their fixed11 dimensions are2 and1. These controls do not determine the actual modular Jacobian's integral lattice.

The individual squarefree mod11 source factor gives a further written argument. For an abelian subvariety B isogenous to the source A, its 11-adic Tate lattice and A's decompose into eight rank-one Z_{11} components under the Hensel idempotents. They are isomorphic. Tate's integral Hom theorem and approximation modulo11 therefore give an actual $A \to B$ isogeny of degree prime to11. Composing $B \to J$ gives a map injective on geometric 11-torsion, whenever J contains the source isogeny factor. CM.md gives our deduction and all hypotheses, using the primary proof in SOURCES.md.

Applied to the smooth compact level77 modular normalization, this uses the frozen written modular/Jacobian bridge. It is an existence result, not an evaluated correspondence or divisor map. The old denominator704 still obstructs the simple coprimality projector; it does not obstruct every integral subgroup-preserving map. Full Tate/abelian formalization remains open; no axiom encoding that conclusion is introduced.

Odd cover: kernel and actual theta lifts

G17 computes U*B*V=diag(1,1,3,105) in17 Smith iterations for the actual odd cover, with explicit two-sided integral inverses for U,V. Thus its split-product kernel is E[3]×E[105], of order99225. Four actual generators have orders3,3,105,105. Independent3/5/7 torsion bases and primitive Weil pairings verify completeness; B kills each generator. Diagonal Theta and 3Theta commutators are all trivial, with actual finite congruences proved in Lean.

trace model	Frobenius orders on3/5/7 torsion	common coordinate field
E0	8,24,42	F_(2^168)
E1	8,24,21	F_(2^168)

The certified field modulus has low bits95. Combining these coordinates with the F32 split-cover identification needs F_(2^840); that identification and F2 compatibility remain unevaluated.

G18 constructs exact factored translation operators for all four kernel generators on the diagonal bundle of degrees[9,63,315,855], including the nontrivial Pic2 corrections. The exported functions use full Miller addition chains and explicit lines through T and-a. Period scalars agree at three independent tuples. Exact3/5/7 roots normalize each operator to its required order; all roots are checked. All18 commutator probes per graph pass. Independent point addition checks the correction-square identity at48 coordinates per graph. A universal integer residual and its rational characteristic2 consequence are proved in Lean.

Under the written Miller/divisor bridge, the constant period and commutator ratios make these normalized operators a geometric kernel linearization. They need not be the canonical character of the original mixed divisor. That matching and F32/F2 descent are still open. The factored averaging operator is an idempotent invariant projector; Lean proves the general statement for any supplied representation of this odd order.

descent stage	section dimension under written geometry
diagonal cover	8704314675
after odd kernel	87723
after19^2 quotient	243

The functions and projector descriptors do not materialize dense matrices or extract an invariant basis. The final243 target sections and proper embedding equations remain unconstructed. THETA.md records exactly which linearization and divisor obligations have been checked.

Saturation: the four requested survivors are excluded

Q23 normalizes actual source cones independently for inputs0,2,4,6. Sixteen quartic separators have membership depth6 in h. The second-factor separator rejects the unit cubic of each actual Q22 model, excluding all four points for arbitrary ramification under the exported source semantics. Generated Lean binds the actual units/gates and checks the scalar separators.

Q24 retains every original and prior repair constraint, adding only the valid radical boundary consequence that all four ordinary cubic nodes vanish. Four new models survive, each found in nine search visits:

graphs	nonzero nodes
0,1	59,61,64,65,66
2,3	58,60,63,64,65

They lie in the supersingular companion subcircuit and are not yet classified as false or genuine closure points. Lean checks their complete finite prior constraints and cubic zeros. Full saturated ideal equality and proper quotient embeddings remain open. SATURATION.md gives the source bindings.

Validation, failures and remaining work

There are69 new named Lean results. The full project build and transitive axiom audit cover553 project theorems with0 failing and0 trusted; VALIDATION.md and the raw gate receipts give the exact scope. Six new JSON outputs and five generated Lean modules have exact replay checks; three new source-bound IDC records cover the correction residual, CM norm product and T translation. PROOF_MAP.md distinguishes kernel statements, finite native checks and written global geometry.

The root release suite passes4487 tests with119 ignored; native tests pass115 with1 ignored. Failed Rust type/serde builds, the stale-binary invocation, and the Lean cast/denominator-clearing attempts remain in raw logs. Earlier G17/G18 outputs are retained before stronger inverse and square-identity controls. No failed run is promoted as evidence.

Next construction obligations: an explicit integral level77 correspondence and its actual subgroup image; canonical mixed-divisor character matching, F32/F2 compatibility and invariant section extraction; a companion separator or full arc for the new Q24 models. Round14 ranks and signed divisors and Round15 norm source-fiber gaps remain frozen. The user retains IC assessment.

New identity certificates: Pic2 correction IDC1h31bc950ebd40dee8; CM norm IDC1h9a53998afa83f290; T translation IDC1h9e3d3e3bc4edfd8a. Their JSON files and exact Lean source hashes sit beside this report. The native ledger pins153 sources and96 raw files, including retained failed attempts.

Evidence links

Companion files: CM.md, THETA.md, SATURATION.md, PROOF_MAP.md, SOURCES.md, ledger.json, runs.sha256, runs.tar.xz and REPLAY.sh. Primary integral Tate theorem source: Sam Lichtenstein, The Tate conjecture, Theorem 1.1 and Section 3, math.stanford.edu/~conrad/mordellsem/Notes/L03.pdf. The existence deduction and all modular/divisor bridges remain written; they are not full Lean geometry proofs.